CLOSED POINTS AND CLOSED-IMMERSIVE RESIDUAL SPACES UNDER LOCAL NOETHERIAN HYPOTHESES

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Editorial note. Counted span is the author's own proof environment, source lines 3458-3470, immediately following the statement at 3450-3456.

Editorial note. Two cross-chapter pointer families (Morphisms of Spaces, Properties of Spaces) are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Closed immersions have closed images on topological spaces. Morphisms of Spaces, Lemma spaces-morphisms-lemma-immersion-when-closed, cited at proof lines 3458-3459. Forward direction: Z_x -> X closed immersion makes x closed in |X|.
- Reduced closed subspace with a given underlying point. Properties of Spaces, Lemma spaces-properties-lemma-reduced-closed-subspace, cited at proof lines 3462-3464. Produces the reduced closed subspace Z with $|Z| = \{x\}$ that the converse identifies with Z_x.
- Closed immersions are locally of finite type; finite type over locally Noetherian is locally Noetherian. Morphisms of Spaces, Lemmas spaces-morphisms-lemma-immersion-locally-finite-type and spaces-morphisms-lemma-locally-finite-type-locally-noetherian, cited at proof lines 3465-3467. Z is locally Noetherian, which is needed to match the definition of the residual space.

Source reference. Original reference 06ED: focused-seeds/geometry/S9-decent-residual-space-closed/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04CD: focused-seeds/geometry/S9-decent-residual-space-closed/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04ZK: focused-seeds/geometry/S9-decent-residual-space-closed/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03IQ: focused-seeds/geometry/S9-decent-residual-space-closed/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted

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proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $x \in |X|$ with residual space $Z_x \subset X$. Assume X is locally Noetherian. Then x is a closed point of $|X|$ if and only if the morphism $Z_x \to X$ is a closed immersion.*

Proof. If $Z_x \to X$ is a closed immersion, then x is a closed point of $|X|$, see Morphisms of Spaces, Lemma 04CD. Conversely, assume x is a closed point of $|X|$. Let $Z \subset X$ be the reduced closed subspace with $|Z| = \{x\}$ (Properties of Spaces, Lemma 03IQ). Then Z is locally Noetherian by Morphisms of Spaces, Lemmas 06ED and 04ZK. Since also Z is reduced and $|Z| = \{x\}$ it $Z = Z_x$ is the residual space by definition. □